

## ENBC321/BIOE658I: Homework #2

**Due:** Monday, 02/16/26 by 5:00 PM, beginning of the class

**Submit your homework to ELMS, including Python code and the dataset**

**Overall points: 60**

**Note:** Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

**Note:** Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

**Note:** Please name your codes using the following format: "problem1\_fullname.py". Start with the problem number and followed by your full name, e.g., "**problem1\_jimmyazarnoosh**".

**Note:** Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of ZERO. Repeated violations may lead to more severe consequences.

**Note:** Submit **a single zip file** containing the following:

- The dataset you used
- Python code files
- Your answers to the questions in PDF format

### Simple and multiple linear regression

In this assignment, you will replicate your previous Statistics assignment, but this time, you will use **Python** instead of R.

**Problem 1.** You are tasked with analyzing data collected from a small bakery to understand the relationship between the number of advertisements they run each month ( $x$ , in units) and their monthly sales ( $y$ , in \$1000). (20 points)

The bakery provides the following data for the last 6 months:

| Month | Number of Advertisements ( $x$ ) | Monthly Sales ( $y$ ) in \$1000 |
|-------|----------------------------------|---------------------------------|
| 1     | 1                                | 3                               |
| 2     | 2                                | 5                               |
| 3     | 3                                | 7                               |
| 4     | 4                                | 10                              |
| 5     | 5                                | 12                              |
| 6     | 6                                | 14                              |

a) **Fit the Model:** Compute the slope ( $b_1$ ) and intercept ( $b_0$ ) of the regression line using the formulas and write the equation of the regression line. (10 points)

- I. Write your own custom function to calculate the coefficients  $b_0$  and  $b_1$ .
- II. Use **Python** function `LinearRegression` to calculate coefficients.

**Note:** You should get similar results in both I and II.

b) **Prediction:** Use the regression equation to predict the sales if the bakery runs 4 and 8 advertisements in a month. (5 points)

c) **Calculate  $R^2$ :** Compute the total sum of squares (SST), the regression sum of squares (SSR), and the residual sum of squares (SSE) to find  $R^2$ . (5 points)

**Note:** Your calculated  $R^2$  should match  $R^2$  reported in the summary. This is how we get summary in Python:

```
x_intercept = sm.add_constant(x) # Add intercept term
ols_model = sm.OLS(y, x_intercept).fit()
summary = ols_model.summary()
print("The p-value is:", summary)
```

**Problem 2.** A hospital is studying the effect of a treatment ( $x_1$ ) and patient gender ( $x_2$ ) on recovery time ( $y$ ) in days. The treatment is categorized as “Placebo” and “Drug”, while gender is categorized as “Male” and “Female”. (20 point)

The dataset is provided in the following:

| Recovery Time | Treatment | Gender |
|---------------|-----------|--------|
| 10            | Placebo   | Male   |
| 8             | Drug      | Female |
| 15            | Placebo   | Female |
| 12            | Drug      | Male   |
| 7             | Drug      | Male   |
| 14            | Placebo   | Female |
| 9             | Drug      | Female |
| 11            | Placebo   | Male   |
| 6             | Drug      | Female |
| 13            | Placebo   | Male   |

```
# Create the dataset
data = pd.DataFrame({
    "Recovery_Time": [10, 8, 15, 12, 7, 14, 9, 11, 6, 13],
    "Treatment": ["Placebo", "Drug", "Placebo", "Drug", "Drug", "Placebo", "Drug",
"Placebo", "Drug", "Placebo"],
    "Gender": ["Male", "Female", "Female", "Male", "Male", "Female", "Female", "Male",
"Female", "Male"]
})
```

a) **Fit the Model:** Fit a linear regression model to predict recovery time based on treatment and gender and write the equation of the model. (5 points)

b) Explain what the coefficients for the treatment and gender dummy variables represent. (7 points)

c) Predict the recovery time for

- a female patient who received the drug. (4 points)
- a male patient who received the drug. (4 points)

**Hint:** Use 0 or 1 in the regression model to calculate recovery time.

**Problem 3.** A researcher is studying the factors affecting house prices in a city. They have collected data on 50 houses with the following variables: (20 points)

**Price:** The selling price of the house (in thousands of dollars).

**Size:** The size of the house (in square feet).

**Bedrooms:** The number of bedrooms in the house.

**Age:** The age of the house (in years).

**Distance\_to\_City\_Center:** The distance of the house from the city center (in miles).

The dataset is provided in the following:

```
# Given dataset
np.random.seed(123)
Price = np.round(np.random.normal(loc=300, scale=50, size=50), 2)
Size = np.round(np.random.uniform(low=800, high=4000, size=50), 2)
Bedrooms = np.random.choice(range(2, 7), size=50, replace=True)
Age = np.random.choice(range(1, 51), size=50, replace=True)
Distance_to_City_Center = np.round(np.random.uniform(low=1, high=30, size=50), 2)

# Create DataFrame
data = pd.DataFrame({
    "Price": Price,
    "Size": Size,
    "Bedrooms": Bedrooms,
    "Age": Age,
    "Distance_to_City_Center": Distance_to_City_Center
})
```

a) **Fit the Model:** Fit a linear regression model to the dataset and write the equation of the regression model based on the output in summary. Use summary in your code to show,  $R^2$ , p-value, etc. (10 points)

b) Which variables significantly impact the house price? Justify your answer based on the p-values. Consider the common significance level of 5%. (5 points)

c) **Prediction:** Predict the price of a house with the following attributes: (5 points)

- Size: 2500 square feet
- Bedrooms: 4
- Age: 20 years
- Distance\_to\_City\_Center: 10 miles